

TransitOps

Smart Transport Operations Platform - Person 2 Fleet, Driver, and Compliance Module

Owner	Akshay Yadav - Person 2
Hackathon	Odoo Hackathon - 8 hour build window
Branch	feature/fleet-compliance
Current repository folder	transitops/
Current detected stack	Backend: Express + Mongoose. Frontend: React + Vite.
Document purpose	Implementation scope, roadmap, integration contracts, and task checklist.

1. Person 2 Responsibility

Person 2 owns the complete Fleet & Driver Management module. Other team members should integrate through stable APIs and shared constants rather than editing this module directly.

- Vehicle Management: CRUD, details, search, filters, sorting, status badges, validation, and retirement.
- Driver Management: CRUD, profile, search, filters, safety score, license validation, and availability.
- Compliance Dashboard: total drivers, valid licenses, expired licenses, expiring soon licenses, average safety score, and suspended drivers.

2. Current Team Foundation

Area	Decision
Backend	Node.js, Express, Mongoose
Frontend	React, Vite
API prefix	/api
Response shape	{ success, message, data } or { success, message, errors }
Shared constants	transitops/backend/src/constants/enums.js
Person 2 branch	feature/fleet-compliance

3. Vehicle Domain

Field	Rules
registrationNumber	Required, unique, trimmed, uppercase, indexed
name	Required vehicle name
model	Optional model
type	Enum from shared vehicle types

Field	Rules
maximumLoadCapacity	Required, must be greater than zero
odometer	Required, cannot be negative
acquisitionCost	Optional, cannot be negative
region	Optional operational region
status	Available, On Trip, In Shop, Retired

4. Driver Domain

Field	Rules
name	Required
licenseNumber	Required, unique, trimmed, uppercase, indexed
licenseCategory	Required
licenseExpiryDate	Required date
contactNumber	Required
safetyScore	Required, 0 to 100
region	Optional operational region
status	Available, On Trip, Off Duty, Suspended

5. Business Rules

- A vehicle can be dispatched only when its status is Available.
- Vehicles with status On Trip, In Shop, or Retired must not appear in dispatch availability lists.
- A driver can be dispatched only when status is Available and licenseExpiryDate is today or in the future.
- Drivers with status On Trip, Off Duty, or Suspended must not appear in dispatch availability lists.
- Expired license drivers must not appear in dispatch availability lists even if their status says Available.
- Duplicate registration numbers and duplicate license numbers must be rejected case-insensitively.

6. Required API Contract

Method	Endpoint	Purpose
GET	/api/vehicles	List/search/filter/sort/paginate vehicles
POST	/api/vehicles	Create vehicle
GET	/api/vehicles/:id	Vehicle details
PUT	/api/vehicles/:id	Update vehicle
DELETE	/api/vehicles/:id	Retire or delete vehicle per final team decision
GET	/api/vehicles/available	Dispatch-safe vehicle list for Person 3

Method	Endpoint	Purpose
GET	/api/drivers	List/search/filter/sort/paginate drivers
POST	/api/drivers	Create driver
GET	/api/drivers/:id	Driver profile
PUT	/api/drivers/:id	Update driver
DELETE	/api/drivers/:id	Delete or deactivate driver per final team decision
GET	/api/drivers/available	Dispatch-safe driver list for Person 3
GET	/api/compliance/drivers	Compliance dashboard metrics

7. Backend Files For Person 2

The following backend files are the intended Person 2 ownership area. Existing shared route and constant files may be touched only to mount routes or add shared enum values.

```
backend/src/models/Vehicle.js
backend/src/models/Driver.js
backend/src/validators/vehicleValidator.js
backend/src/validators/driverValidator.js
backend/src/repositories/vehicleRepository.js
backend/src/repositories/driverRepository.js
backend/src/services/vehicleService.js
backend/src/services/driverService.js
backend/src/services/complianceService.js
backend/src/controllers/vehicleController.js
backend/src/controllers/driverController.js
backend/src/controllers/complianceController.js
backend/src/routes/vehicle.routes.js
backend/src/routes/driver.routes.js
backend/src/routes/compliance.routes.js
```

8. Frontend Files For Person 2

```
frontend/src/features/fleet/vehicles
frontend/src/features/fleet/drivers
frontend/src/features/fleet/compliance
frontend/src/components/common
```

9. Task Checklist

Task	Status	Notes
1	Completed	Inspected initial workspace and found it empty.
2	Superseded	Initial local scaffold was created, then excluded after syncing Person 1 repo.
3	Completed and pushed	Added Person 2 module folders and task map on feature/fleet-compliance.
4	Next	Add Vehicle and Driver Mongoose models.
5	Next	Add validation helpers.
6	Next	Add repositories for database access.
7	Next	Add services with business rules.
8	Next	Add controllers.
9	Next	Add routes and mount them under /api.
10	Next	Add compliance service and endpoint.
11	Next	Add backend tests.
12	Next	Add frontend API functions and UI pages.

10. Git And Team Safety Rules

- Work on branch feature/fleet-compliance.
- Before starting a task, run git fetch origin and confirm branch status.
- Commit only Person 2 files unless route mounting or shared constants are required.

- Avoid git add . because old local root-level scaffold exists outside transitops/.
- Use focused commits such as feat(fleet): add vehicle and driver models.
- Do not edit Person 1 auth/dashboard files, Person 3 trip/maintenance files, or Person 4 finance/report files.

11. Integration Points

Team member	Needs from Person 2
Person 1	Fleet summary metrics for dashboard and RBAC route protection alignment.
Person 3	GET /api/vehicles/available and GET /api/drivers/available for dispatch.
Person 4	Vehicle and driver lists/details for fuel logs, expenses, and reports.

12. Edge Cases To Cover

- Registration numbers and license numbers should be unique after trimming and uppercase normalization.
- Invalid enum values should return validation errors.
- Capacity zero or negative must be rejected.
- Negative odometer and negative acquisition cost must be rejected.
- License expiry today should be considered valid until the day ends.
- Expired license drivers should be excluded from available driver endpoint.
- Pagination should handle empty results and out-of-range pages gracefully.

13. Final Deliverable Standard

Person 2 delivery is complete when vehicle CRUD, driver CRUD, available vehicle/driver APIs, compliance metrics, validations, business rules, frontend management pages, and basic tests are working on feature/fleet-compliance without modifying unrelated team modules.